

An Efficient Framework for Management Strategy Evaluation Grid-Searches

AUTHOR

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1 Abstract

This project introduces a computationally efficient framework for evaluating Harvest Control Rules (HCRs). Initially applied to a single-stock fishery, the framework utilises Gaussian process regression to independently model both catch and risk. Then, by combining the methods of Bayesian history matching, acquisition functions and k-means clustering, we are able to systematically map the parameter space. This drastically reduces the need for expensive simulation runs.

Building on this foundation, we scale up this framework to complex mixed-fisheries scenarios. Here, the objective becomes identifying the fishing mortality for each of the interacting stocks that safely maximizes total long-term catch. This framework provides a powerful tool for marine policy by significantly reducing the computational burden of mixed-fishery MSEs whilst ensuring that the stocks remain firmly within safe biological limits. However, there are limitations to our current work which we discuss throughout and summarise when suggesting further developments.

2 Introduction

Sustainable fisheries management requires a delicate balance of maximising long-term catch whilst keeping the risk of stock depletion below precautionary thresholds ([ICES 2021b](#)). Traditionally, Management Strategy Evaluation (MSE) grid searches are used to find optimal HCRs ([Trusts 2016](#)),([ICES 2019a](#)). However, simulating the complex, interconnected dynamics found in fisheries is computationally expensive and time-consuming, particularly as management shifts from single-stock assessments to more realistic, multispecies mixed fisheries ([Spence 2025](#)),([Sun et al. 2023](#)),([Pace et al. 2025a](#)).

2.1 Key Concepts

We firstly introduce some of the key concepts in fisheries management which will be used throughout this paper. One of these is catch. Unfortunately, there are many different definitions of this term ([ICES 2021b](#)),([Pace et al. 2025a](#)),([United Nations. Fisheries Department 2024](#)). The most commonly used is the ICES definition where catch includes wanted catch, unwanted catch and unaccounted removals ([ICES 2019c](#)). Wanted catch refers to the catch that has commercial value. Unwanted catch refers to the portion of the catch that is thrown back into the sea for being

undersized, having no commercial value or the fleet not having enough quota to land it. Unaccounted removals comes from ICES filling in the gaps in their historical data, and so is not particularly relevant for this project (ICES 2019c),(ICES 2021b). However, some papers may use equations to describe catch which may not take all of these factors into account, notably Pace et al. (2025a) which we will be studying in the second half of this project.

Risk is another key concept. We define risk using the ICES *Prob3* definition as the maximum probability that SSB is below B_{lim} , where the maximum (of the annual probabilities) is taken over a certain number of years (ICES 2019a). There have also been different definitions of this term, but we will use the definition of risk above throughout this paper as it is the version recommended by ICES (ICES 2019a). Throughout this paper, SSB refers to spawning stock biomass which is the weight of adult breeding fish (Lart 2026).

Some other important concepts in fisheries management are fishing mortality and the biomass limit reference point. The fishing mortality of a specific stock, normally denoted as F_{target} , is defined as what proportion of fish are killed due to fishing from that specific stock (Lart 2026). The biomass limit reference point, often denoted as B_{lim} , is defined as the limit where stocks with a spawning stock biomass (SSB) below this level may not have high enough recruitment to sustain a fishery (Lart 2026).

Furthermore, we need to discuss the difference between single-stock and mixed fisheries. Single-stock fisheries consist of only one stock and can be fished by any number of fleets (ICES 2021b). On the other hand, mixed fisheries consist of multiple stocks and can be fished by any number of fleets (Sun et al. 2023). Both types of fishery are normally constrained to a specific geographic area (ICES 2021b). Mixed fisheries are more difficult to manage due to interspecies (trophic) interactions and because gear often catches multiple species at once, which can lead to choke stocks (Sun et al. 2023),(Pace et al. 2025a). Choke stocks are where the amount we are allowed to catch (the quota) of one stock means that the quota for other stocks cannot be met (Sun et al. 2023).

Lastly, we will discuss briefly the ideas of optimisation and Management Strategy Evaluation. In this paper, we use optimisation to find the maximum value of a function over a specific set of parameters (Spence 2025). There are various methods for this when the function is considered as a black-box, some of which we will discuss and use below (Jones, Schonlau, and Welch 1998),(Huang et al. 2006). Management Strategy Evaluation is a tool used in fisheries management to decide whether a specific management strategy (say a HCR) can achieve the pre-agreed management objectives (Trusts 2019). These management objectives may require careful balancing, such as wanting to harvest enough fish to feed a large population whilst protecting the stock for future years, and are given quantitative expressions (Trusts 2019).

2.2 Previous Work and Structure of this Paper

There has been much previous work on this problem. MSEs to test HCRs for single-stock fisheries are a well-established method of fisheries management. They are used by ICES to implement single-stock Total Annual Catches (TACs) and are thus widely adopted (ICES 2019a), (Trusts 2019).

One of the most recent papers on MSEs for single-stock fisheries uses Gaussian Processes to estimate the results of a simulation, due to the simulation being expensive to evaluate (Spence 2025). It then uses Bayesian History Matching to narrow in to the optimal point (Spence 2025). This was helped by previous developments in diverse fields (Jones, Schonlau, and Welch 1998),(Huang et al. 2006),(Frazier 2018).

For many years researchers have realised that a single-stock approach is not sufficient (Ulrich et al. 2012),(Sun et al. 2023). This has led to a growth in the literature relating to how to manage mixed fisheries, which we are building on in Section 4 of our report (Ulrich et al. 2012),(Sun et al. 2023), (Pace et al. 2025a). The MSEs for mixed fisheries tend to be more complicated than for single-stock fisheries as they take more factors into account. Thus, finding ways to improve MSE grid searches in a single stock context and then applying this method to mixed fisheries contexts could be beneficial for the fishing industry (Spence 2025),(Pace et al. 2025a),(Hofert and Mächler 2016). This is our intention throughout this paper. We describe the framework outlined in the abstract in detail with applications to a couple of scenarios, the results of which are discussed in Section 5. We also suggest areas for further development of the method in Section 6.

In the Section 3 part of the project, we build on Spence (2025). We focus on enhancing the methods in the paper by adding acquisition functions and k-means clustering to improve how we choose to sample points between rounds as this was not deeply considered in the original paper (Spence 2025),(Frazier 2018),(Jones, Schonlau, and Welch 1998),(Huang et al. 2006). The aim of the second half of the project in Section 4 is to apply the methods from Section 3 to a more realistic mixed fisheries scenario and to demonstrate how effective our method may be if used in the industry (Frazier 2018),(Pace et al. 2025a),(Sun et al. 2023).

3 Creating the framework in a single stock fisheries context

3.1 General theory

We will look at the mathematics behind the methods in Spence (2025) above and also at the acquisition functions which I have added to the original code. For a comprehensive introduction to the general theory on Gaussian Processes, Maximum Likelihood Estimation and Gaussian Process Regression see Frazier (2018).

Throughout this section, we often call f an objective function as this is the standard across the literature. It should be regarded as separate from our objective function in Equation 8, which clearly needs to be split into separate parts to be emulated by a Gaussian Process (GP). Until the end of Section 3.1, f should be regarded as an objective function we wish to maximise which is emulated by a GP.

3.1.1 Gaussian Processes

A Gaussian Process F has a mean function μ_0 and a covariance function $\text{cov}_0(x_i, x_j)$. We can then evaluate the covariance function $\text{cov}_0(x_i, x_j)$ for every pair x_i, x_j where $i, j \in \{1, \dots, n\}$ to find the covariance matrix $\Sigma_{1:n}$. Then, F is a probability distribution over our objective function f with the property that, for any given collection of points $x_1, \dots, x_n$, the marginal probability distribution on $F(x_{1:n}) = (F(x_1), \dots, F(x_n))$ is given by:

$$F(x_{1:n}) \sim N(\mu_0(x_{1:n}), \Sigma_{1:n}) \quad (1)$$

where

$$\mu_0(x_{1:n}) = (\mu_0(x_1), \dots, \mu_0(x_n))$$

and where [Equation 1](#) can be referred to as the prior distribution of the GP, or simply the prior.

We choose a covariance function such that if for some x, x', x'' in the parameter space we have $\|x - x'\| < \|x - x''\|$ for some norm $\|\cdot\|$, then $\text{cov}_0(x, x') > \text{cov}_0(x, x'')$. We use a GP to emulate the objective function because it is much cheaper to evaluate than our objective function. We can use the GP $F(x)$ as our belief of the distribution of values for $f(x)$ for every point x in the parameter space. This is true even for the evaluated points $x_1, \dots, x_m$ as the GP is fitted to these points ([Spence 2025](#)).

3.1.2 Maximum Likelihood Estimation

When using GPs to emulate our objective function, we need to be able to estimate the hyperparameters of the GP using the data we gain from evaluating our points $x_1, \dots, x_n$. We can do this as follows. Firstly, we let the vector η represent the hyperparameters of the GP. Then, given the observations $f(x_{1:n}) = (f(x_1), \dots, f(x_n))$, we calculate the likelihood of these observations under the prior distribution of the GP given η which is denoted as $p(f(x_{1:n})|\eta)$ and modelled by [Equation 1](#). Lastly, we set $\hat{\eta}$ to the value that maximizes this likelihood:

$$\hat{\eta} = \text{argmax}_{\eta} p(f(x_{1:n})|\eta)$$

3.1.3 Gaussian Process Regression

Gaussian process regression is a Bayesian statistical method for modelling functions. Again, let f be the objective function and we focus on the parameter space $X := \{x_1, \dots, x_k\}$. Now, if we have sampled n points such that we have $f(x_{1:n})$ and are considering sampling x_{n+1} next, we let $k = n + 1$ in [Equation 1](#). Then, we can compute the conditional distribution of $F(x_{n+1})$ given $f(x_{1:n})$ using Bayes' rule:

$$F(x_{n+1})|f(x_{1:n}) \sim N(\mu_n(x_{n+1}), \sigma_n^2(x_{n+1})) \quad (2)$$

where:

$$\mu_n(x_{n+1}) = \text{cov}_0(x_{n+1}, x_{1:n})(\Sigma_{1:n})^{-1}(f(x_{1:n}) - \mu_0(x_{1:n})) + \mu_0(x_{n+1})$$

$$\sigma_n^2(x_{n+1}) = \text{cov}_0(x_{n+1}, x_{n+1}) - \text{cov}_0(x_{n+1}, x_{1:n})(\Sigma_{1:n})^{-1} \text{cov}_0(x_{1:n}, x_{n+1})$$

where:

$$\text{cov}_0(x_{n+1}, x_{1:n}) = (\text{cov}_0(x_{n+1}, x_1), \dots, \text{cov}_0(x_{n+1}, x_n))$$

The distribution $F(x_{n+1})|f(x_{1:n})$ is called the posterior probability distribution. We can calculate this distribution for every point in the parameter space X by changing the value of x_{n+1} . This could lead towards being able to calculate a new GP F_n so that we can update our GP every round based on the new points we have evaluated. However, in our framework we have chosen instead to create a new GP each round based on the current data we have instead of updating our GP from the previous round as above. Despite this, we do use [Equation 2](#) to find $F(x_{n+1})|f(x_{1:n})$ for each point x_{n+1} in the parameter space that we have not already sampled. This then allows us to move on to using BHM as we have probabilities for the value of $f(x_{n+1})$ from the posterior probability distribution.

3.1.4 Bayesian History Matching

For the theory here, please see Spence (2025). Let x be a point in the parameter space. We begin with some uncertainty about our objective function $f(x)$, and want to find it's maximum. We can make probabilistic statements such as:

$$P(f(x) > a) = \int_a^\infty P(f(x))df(x)$$

Once we evaluate another point x_1 where $x_1 \neq x$, we are able to use Bayes' Theorem improve our integral to:

$$P(f(x) > a|f(x_1)) = \int_a^\infty P(f(x)|f(x_1))df(x)$$

We now let $a = \max\{f(x_1), \dots, f(x_n)\}$ where n is the number of points we have evaluated so far. As the rounds increase we make sure to include all previous points of the objective function that have been evaluated. We remove the point x from the list of plausible points if:

$$P(f(x) > a | f(x_{1:n})) = \int_a^\infty P(f(x) | f(x_{1:n})) df(x) < \varepsilon \quad (3)$$

for a small $\varepsilon > 0$ until no plausible points remain. Then, the optimum will be x^* such that:

$$f(x^*) = \max\{f(x_{1:n})\}$$

as our index n counts the number of points we have evaluated throughout the whole simulation.

3.1.5 Expected Improvement

The first type of acquisition function we will look at is Expected Improvement (EI). Suppose we have sampled the points $x_1, \dots, x_n$ and observe the values $f(x_{1:n})$. Then, if we were to return a solution at this point, bearing in mind we observe the objective function f without noise and we can only return points we have already evaluated, we would return $f_n^* = \max\{f(x_1), \dots, f(x_n)\}$ (Frazier 2018). Imagine we then consider evaluating another point x_{n+1} to get $f(x_{n+1})$. We can then define the Expected Improvement as:

$$EI_n(x_{n+1}) := E[(F(x_{n+1}) | f(x_{1:n})) - f_n^*]^+$$

where $[((F(x_{n+1}) | f(x_{1:n})) - f_n^*)^+]$ is the positive part of $[((F(x_{n+1}) | f(x_{1:n})) - f_n^*)^+]$. This acquisition function is relatively easy to optimise and many different methods have been developed for doing this (Frazier 2018).

There is another expression for $EI_n(x_{n+1})$:

$$EI_n(x_{n+1}) = [((\mu_n(x_{n+1}) - f_n^*) \cdot \Phi(Z) + \sigma_n(x_{n+1}) \cdot \phi(Z))^+] \quad (4)$$

where again the notation $[(\cdot)^+]$ means the positive part and where:

$$Z = \frac{\mu_n(x_{n+1}) - f_n^*}{\sigma_n(x_{n+1})}$$

Equation 4 can be gained from Equation 1 by setting $k = n + 1$ and then studying the distribution of $(F(x_{n+1}) | f(x_{1:n})) - f_n^*$. However, we can also consider it as a version of Equation 15 from Jones, Schonlau, and Welch (1998) where we first flip the signs as we are focused on the maximisation case and then set $f_{min} = f_n^*$, $\hat{y} = \mu_n(x)$ and $s = \sigma_n(x)$.

3.1.6 Augmented Expected Improvement

This was included to help make the method perform better for noisy functions which will make it more generally applicable (Huang et al. 2006). To deal with these noisy observations, a change was proposed to the standard EI function as detailed below. This change seems mostly to have been justified by empirical performance (Letham et al. 2018). By adjusting Equation 12 found in Huang et al. (2006) to our own notation, we get that:

$$AEI_n(x_{n+1}) = E[(F(x_{n+1})|f(x_{1:n})) - f_{eb}^*]^+ \left(1 - \frac{\sigma_{obs}}{\sqrt{\sigma_n^2(x_{n+1}) + \sigma_{obs}^2}} \right)$$

where σ_{obs} is the standard deviation of the noise variable set by the user and $\sigma_n(x_{n+1})$ is the standard deviation of $F(x_{n+1})|f(x_{1:n})$ from Equation 2. We have also changed f_n^* to f_{eb}^* which is the highest predicted mean at any sampled point so far so that we take into account that the uncertainty in our observations could cause a large spike (Huang et al. 2006).

3.1.7 Knowledge Gradient

We remove the assumption of EI that we have to return a pre-evaluated point as our best point (Frazier 2018). We also now start by saying that the solution we would choose if we have to stop sampling after n points would be the point in the parameter space with the largest $\mu_n(\cdot)$ value, where $\mu_n(\cdot)$ is the mean vector of the posterior probability distribution after n iterations. We call this maximum x_n^* and then we can say that $F(x_n^*)$ is random under the posterior distribution and has the mean vector after sampling $f(x_{1:n})$ of:

$$\mu_n^* := \mu_n(x_n^*) = \max_x \mu_n(x)$$

where x is any point in the parameter space (Frazier 2018). Then, we imagine that we are now allowed to sample a new point x_{n+1} . We get a new posterior distribution for every point x in the parameter space which we can calculate using Equation 2 by replacing x_{n+1} with x and $x_{1:n}$ with $x_{1:n+1}$ to include our new observation. This will have the posterior mean function $\mu_{n+1}(\cdot)$ defined as:

$$\mu_{n+1}(x) = \mu_n(x) + \frac{\text{cov}_n(x_{n+1}, x)}{\text{var}_n(x_{n+1}) + \sigma_{obs}^2} ((F(x_{n+1})|f(x_{1:n})) - \mu_n(x_{n+1})) \quad (5)$$

where σ_{obs} is a noise variable which can be determined by the user (Ungredda, Pearce, and Branke 2022). The conditional expected value for $F(x_n^*)$ changes to be:

$$\mu_{n+1}^* := \max_x \mu_{n+1}(x)$$

So, we can see that the increase in the conditional expected value of $F(x_n^*)$ by sampling the new point x_{n+1} is:

$$\mu_{n+1}^* - \mu_n^* \quad (6)$$

While this quantity is unknown before we sample x_{n+1} , we can calculate it's expected value given our observations $x_1, \dots, x_n$. The Knowledge Gradient (KG) for sampling at a new point x in the parameter space is defined as ([Frazier 2018](#)):

$$KG_n(x) := E[\mu_{n+1}^* - \mu_n^* | x_{n+1} = x] \quad (7)$$

The easiest way to calculate the KG is via simulation. This can be done by simulating one possible value for $f(x_{n+1})$ and then calculating [Equation 6](#). We iterate this process many times so that we can find the average of $\mu_{n+1}^* - \mu_n^*$ and this allows us to estimate $KG_n(x)$ ([Frazier 2018](#)). This process, or calculating $KG_n(x)$ directly from the properties of the normal distribution, both work well in discrete, low dimensional problems which is the situation we are in for the first half of the project. We would sample the point x with the largest $KG_n(x)$ as our next point ([Frazier 2018](#)).

3.1.8 K-means process for selecting multiple points

Combining a clustering method with an acquisition function was an idea I had early on in the project. I was then able to find literature on the subject, including using k-means clustering. For the general theory on k-means processes and an algorithm for implementing a k-means process, please see Kodinariya, Makwana, et al. ([2013](#)) and Hartigan and Wong ([1979](#)).

Let us have the parameter space X as before. We want to be able to select multiple points to sample in our next round so that we continue the pattern set up in the original paper, whilst keeping a good trade off between exploration and exploitation ([Azimi, Fern, and Fern 2010](#)). So, we use the k-means clustering method, which is the most commonly used due to its simplicity compared to other clustering algorithms.

This algorithm creates k clusters (groups of points) such that the points within each cluster have the sum of squares to the centre of their cluster smaller than it would be to the centre of any other cluster ([RDocumentation 2025](#)). It starts by defining k centroids which should be placed as much as possible far away from each other. Then, we take each point in the space and associate it to the nearest centroid. We stop when every point has been assigned to a centroid. At this point, we recalculate k new centroids as the centers of the clusters created by the previous step. This may result in some points changing clusters. We repeat this process until no points change clusters.

However, before the algorithm can start, we must specify how many clusters we want. This can be difficult in many cases. In our case, it is relatively simple as we know how many points we want to sample next and so we set this to be the number of clusters. Lastly, we run the algorithm on X to form the clusters and then we pick the point with the highest value of the acquisition function from each cluster to sample in our next round.

3.2 Application of theory in this half of my project

3.2.1 Context

We have the objective function [Equation 8](#) in Spence (2025) which combines the risk and the catch and so this is the function we want to maximise to get our maximum median long-term catch with the constraint of $risk \leq 0.05$. To do this, Spence (2025) takes a Bayesian History Matching (BHM) approach. We slightly modify this approach. Firstly, we sample eight points which are spaced evenly throughout the parameter space to get some initial data. Then, for each round we do the following:

- We set up or update the Gaussian Process (GP) to model $\ln(risk)$ and the GP to model $\ln(median\ long-term\ catch)$
- We use the GP modelling $\ln(risk)$ to get the value for the risk at each point in the parameter space
- We then use the risk as a threshold so that we only consider the values of F_{target} and $B_{trigger}$ that have a high probability of having risk below 0.05
- We use the GP which is modelling $\ln(median\ long-term\ catch)$ to get the value for the median catch at every point in the parameter space, which will have some uncertainty
- We use BHM to remove any points that are implausible (that have a low probability of being higher than the current best median long-term catch)
- We create eight clusters and select the point with the highest value of the acquisition function from each cluster. These are the eight points we sample in the next round.

We are sampling eight points per round to reduce computation time ([Shahriari et al. 2016](#)), ([Spence 2025](#)). This can be increased if we increase the number of cores we have available, as seen in the second half of this paper ([Shahriari et al. 2016](#)).

3.2.2 Set up the Gaussian Processes

This process was described briefly in Spence (2025) and I repeat it here in more detail to help the reader. We are focusing on maximising the objective function from the paper which is given below:

$$g(\theta) = I_{[0.95,1]}(P(B(\theta) > B_{lim})) \times C(\theta) \quad (8)$$

where for a parameter:

$$\theta = (F_{target}, B_{trigger})$$

in the parameter space we have that $C(\theta)$ is the median long-term catch, $B(\theta)$ is the long-term SSB and:

$$I_{[0.95,1]}(x) = \begin{cases} 1 & \text{if } x \in [0.95, 1] \\ 0 & \text{otherwise} \end{cases}$$

is an indicator function. Here, Spence (2025) defines *risk* as:

$$risk = P(B(\theta) < B_{lim}) \quad (9)$$

Hence, this indicator function tells us if $1 - risk \geq 0.95$ and so if $risk \leq 0.05$ which is what we desire for our precautionary threshold.

To maximise the objective function, Spence (2025) uses two GPs where the risk GP models $\ln(risk)$ and the catch GP models $\ln(C(\theta))$. Maintaining the notation from Spence (2025), we use m_1 for the mean function of the catch GP and m_2 for the mean function of the risk GP:

$$m_1(\phi) = \beta_{1,0} + \beta_{1,1}(\ln(\phi_1 + 0.1)) + \beta_{1,2}(\ln(\phi_1 + 0.1))^2 + \beta_{1,3}(\ln(\phi_1 + 0.1))^3 + \beta_{1,4}(\phi_2 \ln(\phi_1 + 0.1)) + \beta_{1,5}\phi_2 \quad (10)$$

$$m_2(\phi) = \beta_{2,0} + \beta_{2,1}\phi_1 + \beta_{2,2}\phi_2 + \beta_{2,3}\phi_1\phi_2 \quad (11)$$

where all of the above $\beta_{s,t}$ for $s \in \{1, 2\}$ and $t \in \{1, 2, 3, 4, 5\}$ are coefficients to be found through maximum likelihood estimation and:

$$\phi_1 = \frac{F_{target} - 0.1}{0.4} \quad \text{and} \quad \phi_2 = \frac{B_{trigger} - 110000}{90000}$$

as we have rescaled for numerical stability in the GP. Our covariance function c for both GPs is the variance σ_i^2 (which is acting as a scalar) times the Ornstein-Uhlenbeck correlation function:

$$r_i(\phi, \phi', \delta_i) = \exp \left(-\frac{|\phi_1 - \phi'_1|}{\delta_{i,1}} - \frac{|\phi_2 - \phi'_2|}{\delta_{i,2}} \right) \quad (12)$$

where $\delta_{i,1}$ and $\delta_{i,2}$ are the length scales for each of the terms respectively (Williams and Rasmussen 2006).

We need to sample our first round of eight points before setting up the GPs so that we have enough data to estimate all of the coefficients in our GPs (Jones, Schonlau, and Welch 1998). Note that until

we have sampled sixteen points, we set the prior of the catch GP to be the same as the risk GP, $m_2(\phi)$, except with different coefficients (Spence 2025). This is because we need to estimate the coefficients for the mean function for the catch GP and the length scales for the covariance function c from the same data (Jones, Schonlau, and Welch 1998),(Roustant, Ginsbourger, and Deville 2012a). However, after our first round we set the mean function for the catch GP to $m_1(\phi)$ (Spence 2025).

We can set up Gaussian Processes (GPs) as described above in R using the DiceKriging package and use maximum likelihood estimation to get the hyperparameters of $m_1(\phi)$, $m_2(\phi)$ and c for our GPs (Roustant 2025),(Roustant, Ginsbourger, and Deville 2012a). These GPS are as below:

Gaussian Process	What it models
$LC(\theta)$	The natural log of the long-term median catch, i.e. $\ln(C(\theta))$.
$LR(\theta)$	The natural log of <i>risk</i> , where <i>risk</i> is as in Equation 9.

Then, we use $LC(\theta)$ and $LR(\theta)$ respectively to predict $\ln(C(\theta))$ and $\ln(risk)$ at every point in the parameter space. We can then exponentiate these results where needed. We are building our GPs with $\ln$ of the values we want because this helps us generate better predictions (Huang et al. 2006). We have also been able to visualise these GPs in some of the code linked in Section 7.

3.2.3 Excluding implausible points

Recall from Section 3.2.2 that we have the GP $LC(\theta)$ which is modeling the $C(\theta)$ part of the objective function. As the first half of the objective function f is an indicator function, we can focus on maximising $C(\theta)$ whilst meeting the threshold stated by the indicator function (Frazier 2018), (Jones, Schonlau, and Welch 1998).

Firstly, we enforce the threshold in the indicator function by calculating $P(LR(\theta) \leq \ln(0.05))$, which is equivalent to the precautionary threshold $P(risk \leq 0.05)$. We calculate this by making predictions for $\ln(risk)$ at each point using $LR(\theta)$ and finding the probability that these predictions are below $\ln(0.05)$. This probability can be found using Equation 2. Then, we exclude any points with $P(LR(\theta) \leq \ln(0.05)) \leq \varepsilon = 0.0001$ by setting their value for the acquisition function equal to 0. This means they will not be chosen as a point to sample in the next round.

To maximise $LC(\theta)$, we use Bayesian history matching to speed up the process by removing any points that are implausible. We do this by setting $LC(\theta) = f(x)$ and letting a be the natural log of our current maximum long-term median catch, both in Equation 3. We then carry out the process described in Section 3.1.4. This full process ensures that only points that meet the precautionary threshold and are plausible according to Equation 3 can be selected.

3.2.4 Deciding on the next point to sample

We then use acquisition functions in conjunction with BHM. This is partially supported by the literature and was in part an experiment to see if these methods working together helped to speed up the process compared to that in Spence (2025).

In the literature, this method can be understood through the lens of Constrained Bayesian Optimisation ([Letham et al. 2018](#)),([Shahriari et al. 2016](#)). We are using BHM to determine which points are possible based on the constraints:

$$P(C(\theta) > C(\theta^*)) > 0.0001 \text{ and } P(risk \leq 0.05) > 0.0001$$

where θ, θ^* are points in the parameter space and θ^* gives our current maximum long-term median catch as $C(\theta^*)$. The points meeting these conditions will be called the Possible Space, PS . Then, within PS our acquisition function searches for the optimal parameters ([Letham et al. 2018](#)). It is important to reassess these constraints each round because as we gain more information different sets of points may be included in PS ([Letham et al. 2018](#)),([Williams and Rasmussen 2006](#)).

This was an interesting experiment to me because I wanted to explore my hypotheses that:

1. Removing the implausible points using BHM would help the acquisition function focus on testing plausible points
2. Using the acquisition function to sample in areas that are likely to be optimal would help the BHM to be able to remove more implausible points early on

These hypotheses originated from reading the general literature, but specifically Shahriari et al. ([2016](#)), Letham et al. ([2018](#)), Spence ([2025](#)) and Frazier ([2018](#)).

We have three different acquisition functions we have investigated using here. For Expected Improvement, we use the [Equation 4](#) expression and calculate the EI for every point in the parameter space. For Augmented Expected Improvement (AEI), as our situation has no noise, in our code we are still using f_n^* ([Spence 2025](#)),([Frazier 2018](#)),([Huang et al. 2006](#)). Thus, our equation for AEI becomes the same as for EI. For Knowledge Gradient, we estimate the KG from 100 simulations and update the KG each round using [Equation 5](#).

We have now been able to determine which points are possible and have then assigned a value to each point in PS using one of the acquisition functions above. Now, we want to decide which 8 points are best to evaluate next. We use the `kmeans` function which is part of the stats package in R to do this. This function uses the algorithm from Hartigan and Wong ([1979](#)) by default and thus implements our k-means algorithm described in [Section 3.1.8](#). We set $k = 8$ to create eight clusters so that we select eight points to sample next round. This method allows us to search for viable points by looking in the PS but also to keep the points we are going to sample spread out so that we can balance exploitation and exploration more effectively ([Azimi, Fern, and Fern 2010](#)).

3.2.5 Updating our GPS

In our second round, we need to update our GPs with new data ([Spence 2025](#)). We do this by adding in the data for the points that have been newly evaluated this round and creating new GPs from scratch using this, as described in [Section 3.1.3](#).

Now, we repeat the full process described in [Section 3.2](#) until there are no points with an acquisition function value greater than zero. Then, the optimal point is the x^* that has the highest median catch. In our case, x^* is the F_{target} and $B_{trigger}$ that will give the highest median long-term catch whilst following the precautionary principle ([Spence 2025](#)).

3.3 Experimenting with the kernel and mean

For an in depth discussion of the different properties and uses of different kernels and means, see Williams and Rasmussen ([2006](#)). This source is the basis for the majority of this section.

Along with my project supervisor, I decided to run some experiments determining whether changing the kernel and mean used in the Case Study in Spence ([2025](#)) would have an effect on the value determined as the optimal point. This is because the smoothness of the function emulated by the GP can have an impact on what kernel we may want to use in the GP. In [Section 3.2](#), we used the exponential kernel, also called the Ornstein–Uhlenbeck correlation function, as in Spence ([2025](#)). We have experimented below with using the Gaussian kernel (general form shown in [Equation 13](#)) or the Matern kernel with $\nu = \frac{5}{2}$.

$$r(\phi) = \exp\left(-\frac{r^2}{2l^2}\right) \quad (13)$$

The mean was also identified as a possible issue. By assuming a very specific mean (shown in [Equation 10](#) or [Equation 11](#)) for the GP we are assuming that we know a lot about the objective function f . Thus, it was suggested that I try using a zero mean which is commonly done when we do not want to make any assumptions about our objective function.

3.3.1 Experimenting with the kernel

The assumption made by the Gaussian kernel of infinite differentiability can be too strong for real-world processes as it assumes the function it is emulating is very smooth. Hence, the GP will create a very smooth surface due to this assumption in the Gaussian kernel. If a new point we sample does not fit this assumption, then the GP is unsure how to proceed and so sets the variance very high as if it knew nothing, sometimes called a variance explosion. This results in many points that had been deemed implausible returning to PS and this is likely to continue unless different sampling points are chosen. It is instead recommended that we use a kernel from the Matern class of kernels, which includes the exponential kernel.

My experiments appear to back up this theory. Using the exponential kernel, the GP converges to a solution after seven rounds. After changing the kernel to be the Gaussian kernel or the Matern kernel with $\nu = \frac{5}{2}$ it can be seen that neither of these alternatives converge after seven rounds. In each heatmap below, we are plotting how likely the model thinks it is that this point will be a solution with $C(\theta) > C(\theta^*)$, but still precautionary. Any areas with:

$$P(C(\theta) > C(\theta^*)) < 0.0001 \text{ or } P(risk \leq 0.05) < 0.0001$$

have been ruled out as implausible and are displayed as dark blue.

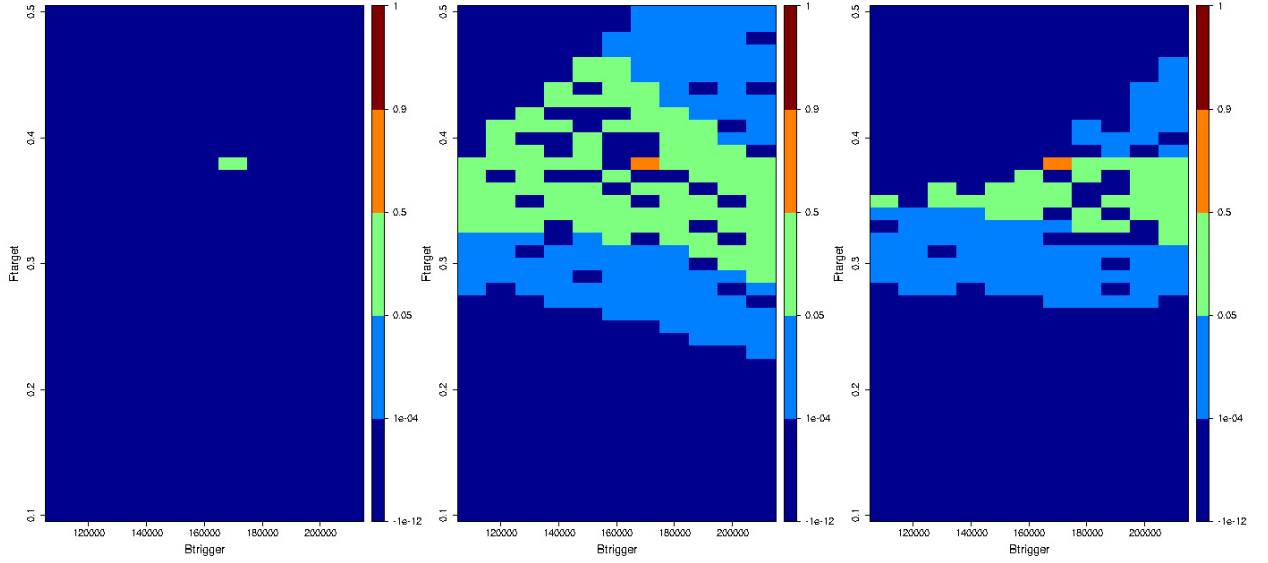


Figure 1: Comparison of convergence under different kernels in the seventh round in case_study8. From left to right: the Exponential kernel, the Gaussian kernel and the Matern 5/2 kernel.

Similar results occur when we run the method once it has been augmented by an acquisition function. As we then allow the method to continue until it has converged, we find that the different kernels cause convergence in a different number of rounds. For example, the EI method converges after seven rounds with the exponential kernel, but after changing to the Gauss kernel it converges after twenty five rounds. When using the Matern 5/2 kernel, the GP does not converge to the correct point.

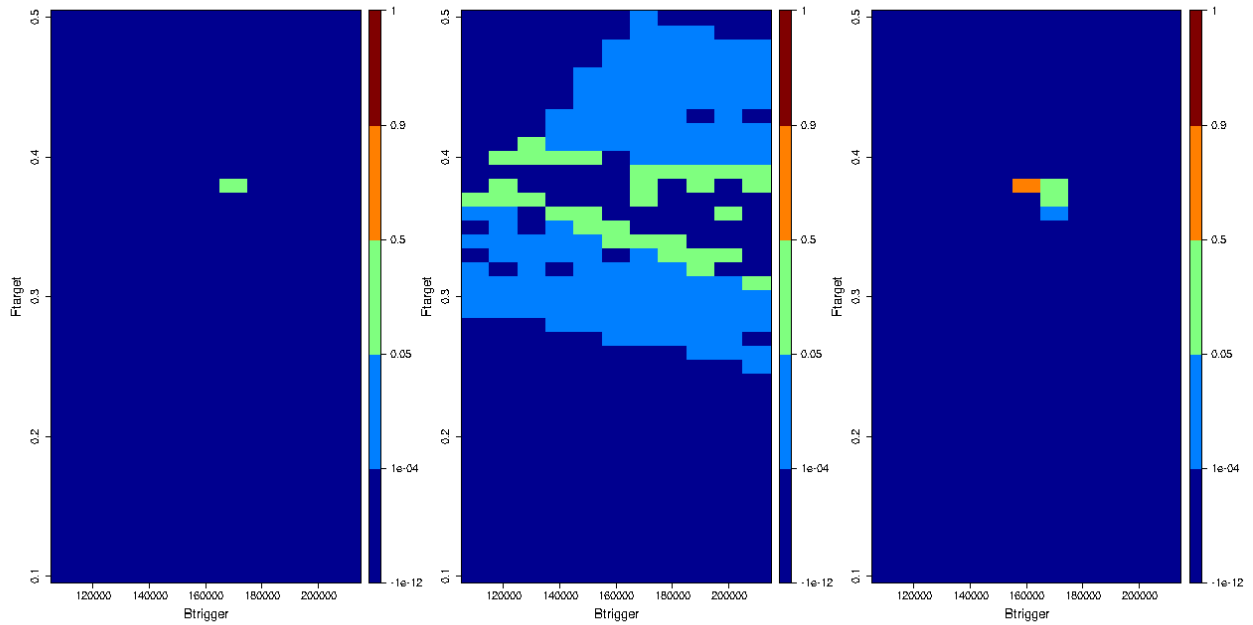


Figure 2: Comparison of convergence under different kernels in case_study8 augmented with the EI acquisition function. From left to right: the Exponential kernel, the Gaussian kernel in the seventh round and the Matern 5/2 kernel in the last round.

3.3.2 Experimenting with the mean

Setting the mean of the GP to zero is more often done in examples to showcase the key concepts than when implementing these methods in real-world situations. My experiments seem to back up this theory as they again show that results are the same or worse than our original approach.

We can see from the heatmaps below that the simulations with means changed to zero either converge at the same round or later on than when we use the original mean specified in ([Spence 2025](#)). This is likely to be because we are able to encode information that we already know about how fisheries and stocks operate when using the original mean, which helps the GP to narrow down to the optimal point in less rounds ([Spence 2025](#)).

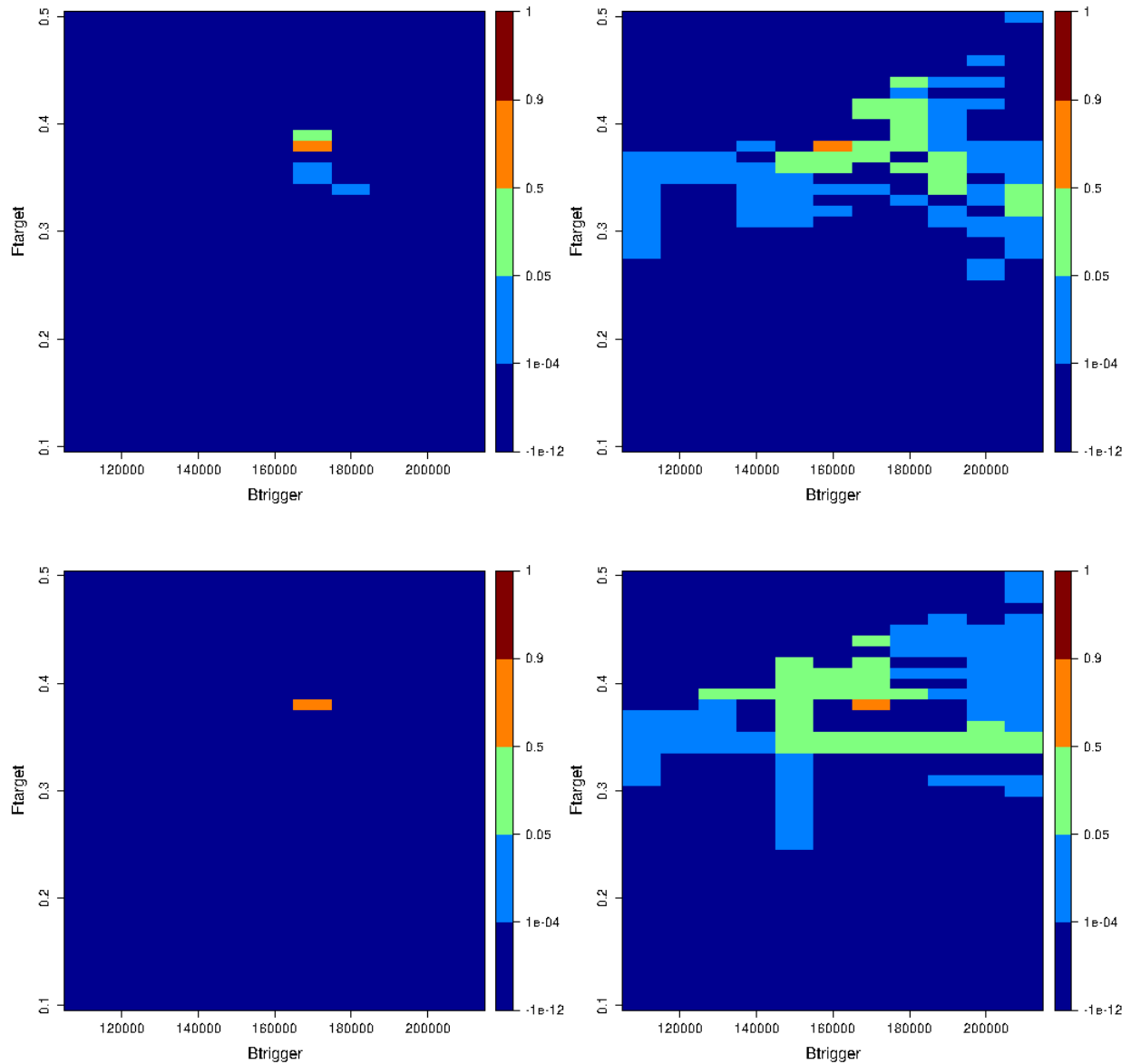


Figure 3: Comparison of convergence under different priors in the seventh round. In the top left, case_study8 with the mean of the GP set to zero only in the first round. In the top right, case_study8 with the mean of the GP set to zero in all rounds. In the bottom left, case_study8 augmented with the KG acquisition function with the mean of the GP set to zero in the first round only. In the bottom right, case_study8 augmented with the KG acquisition function with the mean of the GP set to zero in all rounds

3.4 Determining the best acquisition function

This was done to help us determine which acquisition function was most effective in our specific case as well as highlighting which acquisition function may be most effective in general, which could help us decide which will be most effective in the mixed fisheries context. However, the landscape of the objective function may be different in our mixed fisheries situation due to interactions between different stocks and particularly any choke stocks (Sun et al. 2023), (Pace 2024). This could mean that the acquisition function we choose for the single stock case may not be

the most effective acquisition function in the mixed fisheries case ([Shahriari et al. 2016](#)). However, for now we will use the most effective function in a single-stock context in our mixed fisheries context.

We ran a timing test, a total evaluations test, a number of rounds test and a convergence test comparing the base method used in the Case Study in Spence ([2025](#)) to the base method augmented with the different acquisition functions we have mentioned above. The tests and their results can be seen through the links provided in [Section 7](#).

Despite taking more time on average than the other methods in this case, the KG acquisition function method will be the best to use here because it requires less evaluations of the objective function and so is less expensive to run ([Frazier 2018](#)), ([Jones, Schonlau, and Welch 1998](#)), ([Frazier, Powell, and Dayanik 2009](#)). However, when sampling eight points per round it had the same number of rounds as the other methods for the Case Study. This is important because we evaluate the new points each round in parallel and so KG didn't have the overwhelming lead in terms of computation time that would be expected for less evaluations ([Shahriari et al. 2016](#)). The results from the convergence test are essentially irrelevant for this comparison as all methods got the same result. Due to this decision, we will use KG as our preferred acquisition function throughout the mixed fisheries section of this report.

4 Applying the framework to a mixed fisheries context

4.1 Context

We base the second half of the project around Pace et al. ([2025a](#)), which presents a software package which can be used to project fisheries into the future and is designed to study mixed fisheries. This allows us to run Management Strategy Evaluations for different HCRs that we will parameterise with the F_{target} s of each species we are considering. We optimise based on the catch and ensure that risk is at or below the precautionary threshold. I use the methods from the first half of the project to perform this optimisation.

To be able to simulate the fishery into the future, we use historic data to set up some of the parameters for our model ([Pace et al. 2025a](#)). I focused on the datasets from the Fixed fishing mortality management strategy example and the Exploring simulation outputs example which are both in the MixME documentation ([Pace 2024](#)). However, for the second dataset I was given a shortcut method by a researcher at Cefas which takes a more direct approach to the projection. The code for these datasets is provided in [Section 7](#). Both of these datasets have two stocks (cod and haddock) and two fleets ([Pace 2024](#)). This allows me to use the same methods as in the first half of the project by replacing F_{target} with F_{cod} and $B_{trigger}$ with F_{had} , where F_{cod} and F_{had} are the fishing mortalities for cod and haddock respectively. I was told by the same researcher at Cefas that the stocks are North Sea cod and Celtic Sea haddock, but using citations I can only back up that they are Atlantic cod and haddock ([Pace 2024](#)), ([ICES, n.d.b](#)), ([ICES, n.d.a](#)).

We create our new parameter space to be a grid with all the combinations of F_{cod} and F_{had} both ranging from 0 to 0.6 in 0.02 increments (Spence 2025). This range ensures that we are able to model stock collapse for cod by modelling values above F_{lim} and that we can model stock recovery for cod by including very low values (ICES 2019c). It also allows us to model unsafe fishing for haddock by modelling values above F_{MSY} (ICES 2019b). Recalling that cod is the choke stock for both of our datasets, this parameter space is appropriate (Pace 2024),(Sun et al. 2023),(ICES 2019a).

Due to the change in stocks being modelled, I have decided to keep the prior for the catch GP modelled in the same form as in Equation 11. Despite the more complicated prior used in later rounds for the catch GP in Spence (2025) being designed to approximate yield curves, it may not be appropriate in a mixed fisheries context where the catch of one species is affected by the catch of another (Spence 2025),(Pace et al. 2025a),(Ulrich et al. 2012). Leaving the GP more general avoids mis-specification, ensuring that the GP can be appropriately fitted to the points (Williams and Rasmussen 2006).

For this half of the project, we have switched to modelling SSB directly instead of calculating the risk. We will model $\min(SSB)$ for each stock, which is the minimum value for the SSB of that stock over the years 2030-2039. This is due to the simulation being deterministic as both datasets only have one iteration and the noise for this is pre-calculated (Pace et al. 2025b). These conditions simplify the simulation but mean that we cannot use the standard ICES definition of risk (Pace et al. 2025b),(ICES 2021b). Instead, we adopt another process explained in Section 4.3. We claim that this method satisfies the ICES precautionary standard (ICES 2021b).

In summary, we have the variables below which are important:

Variable	Explanation
F_{cod}	The fishing mortality for cod. Ranges from 0 to 0.6 in 0.02 increments.
F_{had}	The fishing mortality for haddock. Ranges from 0 to 0.6 in 0.02 increments.
$\min(SSB)$ for each stock	The minimum value for the SSB of that stock over the years 2030-2039.

We now can conduct 20 year projections into the future using the MixME package (Pace et al. 2025a). The 20 year projection from 2020-2039 follows guidelines from ICES to create long-term projections based on the biology of the stocks (ICES 2019a). We also follow ICES guidelines to only calculate catch and risk for the last ten years for long-term projection, to allow time for a recovery period (ICES 2019a).

4.1.1 The Optimisation Framework

The process for the first half of the project adapted to our new situation is outlined below. Firstly, note that we are now seeking to maximise the objective function below:

$$h(\theta) = I_{[0,0.05]}(P(\min(SSB_{Cod}) < Cod B_{lim})) \times I_{[0,0.05]}(P(\min(SSB_{Haddock}) < Haddock B_{lim})) \times C(\theta)$$

where the indicator function for cod is:

$$I_{[0,0.05]}(P(\min(SSB_{Cod}) < Cod B_{lim})) = \begin{cases} 1 & \text{if } P(\min(SSB_{Cod}) < Cod B_{lim}) \in [0, 0.05] \\ 0 & \text{otherwise} \end{cases}$$

and similarly for haddock. $Cod B_{lim}$ and $Haddock B_{lim}$ are the B_{lim} for cod and haddock respectively and SSB_{Cod} and $SSB_{Haddock}$ are the SSB for cod and haddock respectively. $h(\theta)$ follows intuitively from the case in [Equation 8](#) once we consider that we are now modelling for multiple stocks and want to maximise the catch whilst keeping both stocks safe.

Firstly, to get some initial data, we randomly sample our first set of $2n - 1$ points, where n is the number of cores the system we are on has. This is again done to reduce computation time ([Spence 2025](#)), ([Shahriari et al. 2016](#)). Then, for each round we do the following:

- We set up or update the GPs to model the $\min(SSB)$ for each stock and the GP to model the total catch from 2030-2039
- We can then use the B_{lim} for each stock as a threshold so that we only consider the values of F_{cod} and F_{had} that have $P(\min(SSB) < B_{lim}) \leq 0.05$
- We use the GP which is modelling the total catch to predict the value for the total catch at every point in the parameter space, which will have some uncertainty
- We use BHM to remove any points that are implausible (that have a probability less than 0.0001 of being higher than the current best total catch)
- We use the KG acquisition function to select $2n - 1$ plausible points to sample in the next round
- We repeat this process until there is only one plausible point left and then we will accept this as being the F_{cod} and F_{had} that maximise the catch whilst keeping $risk \leq 0.05$

Hence, we have the GPs below:

Gaussian Process	What it models
$\min(SSB_{Cod})$	The minimum SSB for cod over the years 2030-2039.
$\min(SSB_{Haddock})$	The minimum SSB for haddock over the years 2030-2039.
$LCM(\theta)$	The natural log of the total catch of cod and haddock given by the MixME model in the years 2030-2039.

4.2 The basics of the MixME model

The vast majority of this description is drawn from Pace (2024), but I repeat it here to ensure description of my specific case. More in depth statistical details can be found in Pace et al. (2025a) and its accompanying supplementary materials.

4.2.1 Starting F_{cod} and F_{had}

We chose our first point to sample as $F_{cod} = 0.28$ and $F_{had} = 0.353$ based upon the values given in the Fixed fishing mortality management strategy example from the MixME wiki. These were drawn from North Sea Cod and Celtic Sea Haddock advice (ICES 2021a),(ICES 2020b). We still sampled $2n - 2$ other points in the first round as well.

4.2.2 Creating the Operating model

We first create the operating model which contains the true data for the stocks and fleets in the dataset. We have data for North Sea Cod from 1963-2019 and for Celtic Sea Haddock from 1993-2019. For the stocks, this includes the numbers, natural mortality, stock mean individual weight and proportion mature split by age and for fleets this includes landing numbers, landing mean individual weight, discard numbers, discard mean individual weight, selectivity and catchability.

In both of our files, this is loaded with the dataset. For every year, the dataset has the catch (in terms of landings and discards) from each fleet and the survivors from that year. It also contains a stock-recruitment model and recruitment is done at the beginning of each time step. The steps for fully assembling the operating model for input into the MixME model are explained below.

4.2.2.1 Estimate historic quota-share for the two fleets

Firstly, we use the operating model to determine the quota share of the catch for each fleet for each year in our projection. This is done by assuming in each fleet that the quota share for each stock corresponds to the proportion of landings of that stock by that fleet.

4.2.2.2 Project structures 20 years into the future

To carry out our 20 year projection, we need to extend the stock and fishery structures forward from 2019 into 2039. There are three categories of parameters that are not estimated dynamically and so need to be extended. These are:

1. All stock parameters, landings and discards mean individual weights and landed fraction
2. Catchability and catch selection
3. Quota-share

We project the parameters in each of these categories from 2019-2039 using the average from the last three years. This means we fill in the actual values we will use, instead of calculating a percentage that is then used in a further calculation like in the quota share section.

4.2.2.3 Calculate numbers of both stocks in initial year

To be able to do our projections of catch, we need to know the number of each stock in each age class at the beginning of the first projection year, 2020. This requires us to do a 1-year short term forecast from our starting point of 2019 using the FLasher package. We set an arbitrary forecast target for this forecast. This is because the forecast target is designed to tell the simulation what to aim for by the end of 2020, but we want the stock numbers at the beginning of 2020 and so this has no effect on the result we want.

4.2.2.4 Creating the Observation Error Model

This is another important component of the model. We create the observation error model where we apply pre-sampled noise to the catch from each fleet. We generate future stock and management advice from this object.

4.2.3 Creating the input object, running the simulation and getting the results

We then use the operating model to make the MixME input object which we can use to run the simulation. This is created from the operating model and the observation error model. Then, we run the simulation. It is important to note that the F_{cod} and F_{had} we choose at the start remain constant throughout the simulation.

The tracking object produced by the simulation contains the catch and the SSB from each year of the simulation, which is how we have been extracting these throughout this document. It also contains simulation performance and diagnostics statistics. We can check for management advice failure, effort optimisation failure and the message given if there was any failure. Effort optimisation failure can be used to see if we are over-fishing the stocks to a point that they will go extinct. Other results we can see are the over-quota catches and the quota uptake. Lastly, we can check which stock is the choke stock.

4.2.4 Additions or Differences in the Shortcut Model

We use the same data as in the Exploring simulation outputs example from the wiki. There are some small differences in how we calculate the HCR and do our one-year forecasts in the shortcut model. However, this is not yet up on the wiki and so we will not be describing it in detail.

4.3 Calculating the risk

MixME defines risk as the proportion of iterations in each year where SSB falls below B_{lim} ([Pace 2024](#)). However, due to only having one iteration in each of my datasets, I have experimented with using a different method to measure the risk ([Pace 2024](#)). In contrast to the first half of the project, we now calculate the risk using the SSB . Note that we also need the B_{lim} for this method. We have a B_{lim} for each stock, taken from ICES advice ([ICES 2020b](#)), ([ICES 2020a](#)). The B_{lim} for cod is 107,000 tonnes and the B_{lim} for haddock is 9,227 tonnes.

Throughout this section, we have assumed that the SSB for each stock for each year outputted by the simulation is the true value. In reality there may be some uncertainty about this value and so we

would need to take this into account (Pace et al. 2025a). This would still allow $\min(SSB)$ to be modelled as a GP and the risk calculation to be the same so that the rest of the process can be carried out as in [Section 4.1.1](#) (as the KG acquisition function can handle noisy situations) (Frazier 2018). However, this is beyond the scope of my project and so I have not implemented this and will not discuss it in detail.

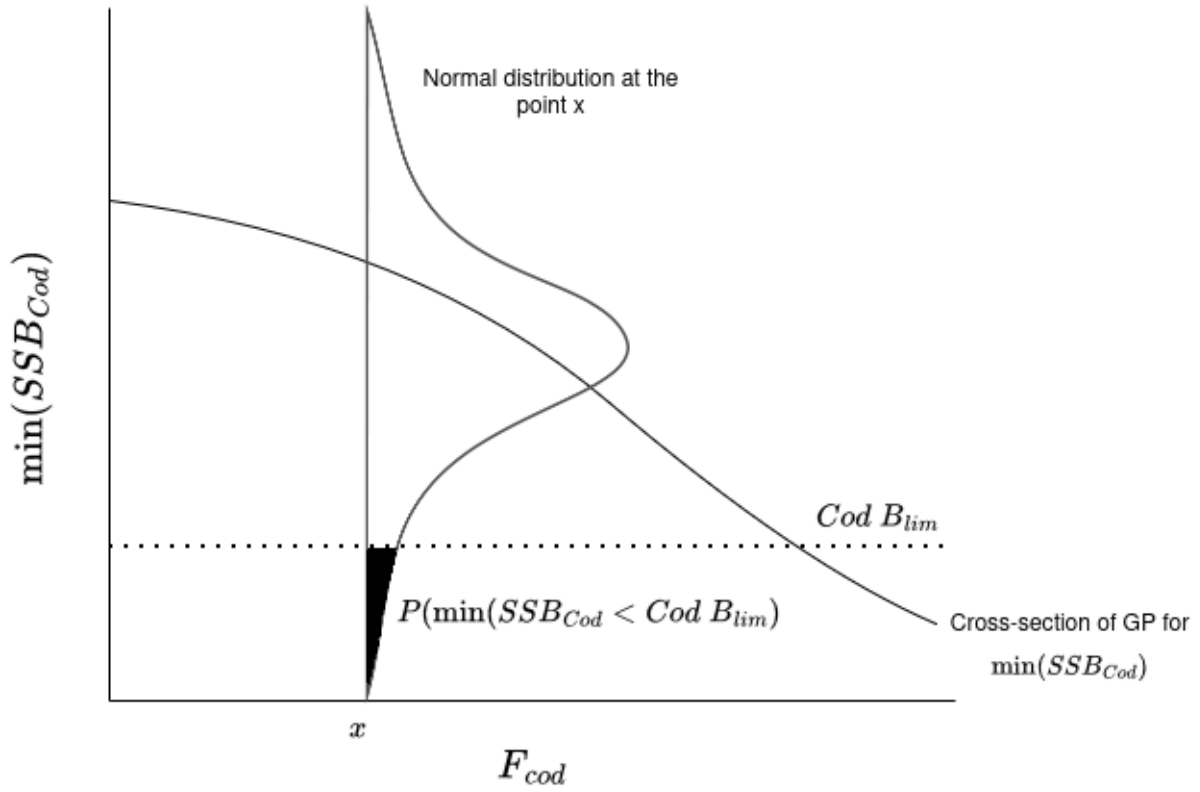
The mean functions of the GPs $\min(SSB_{Cod})$ and $\min(SSB_{Haddock})$ have remained modelled as in [Equation 11](#) from the first half of the project to avoid mis-specification (Williams and Rasmussen 2006),(Roustant, Ginsbourger, and Deville 2012b). The GPs have kept the same estimation method because Maximum Likelihood Estimation is one of the standard methods used in the literature for our situation (Williams and Rasmussen 2006),(Frazier 2018). We also keep the small nugget term as our simulation is still deterministic and we still need our matrices to be invertible (Roustant 2025),(Williams and Rasmussen 2006),(Frazier 2018).

The $\min(SSB)$ for each stock is not necessarily smooth. Due to our discrete time steps, overfishing or low recruitment can cause sudden drops (Pace et al. 2025b),(ICES 2021b). These cannot easily be modelled by any of the kernels we have previously explored due to the size of these drops (Williams and Rasmussen 2006). However, to maintain consistency between the first and second halves of this project and as $\min(SSB)$ is on a similar scale to $C(\theta)$, we choose to continue using the exponential kernel for the GPs in this half of the project.

Firstly, we extract the $\min(SSB)$ for each stock from the result of each of the simulations we have run (Pace 2024),(Pace et al. 2025a). We then put these results into the GPs $\min(SSB_{Cod})$ and $\min(SSB_{Haddock})$. Next, for each stock we predict the $\min(SSB)$ for every point in the parameter space and calculate:

$$risk = P(\min(SSB) < B_{lim})$$

This can be done using [Equation 2](#). A diagram is provided below to improve visualisation, where cross-section means we are taking the GP for a specific value of F_{had} :



Then, we set the KG of any unsafe points to be zero so that they will not be chosen as points to be sampled in the future.

By ensuring for both stocks that:

$$P(\min(SSB) < B_{lim}) \leq 0.05 \quad (14)$$

we ensure that:

$$P(SSB < B_{lim}) \leq 0.05$$

for every year in the long term forecast. This guarantees that the maximum annual risk in this period remains below the 0.05 threshold, which is exactly what is required to meet the ICES precautionary standard due to their definition of *Prob3* (ICES 2019a). This means that this risk calculation could be used in policy documents to set official catch limits in countries that have agreed to this standard (ICES 2019a).

4.4 Parallelisation

This process has been set up so that it is easy to run in parallel. We are able to load the dataset we need once and then run the simulations for each point we have chosen to sample in the round in parallel. As they run, each process only gathers the values we need and removes the rest of the

results of the simulation to save memory. Here, the values we need are the $\min(SSB)$ for cod, the $\min(SSB)$ for haddock and the total catch of cod and haddock over the years 2030-2039. Then, we collect these results and set up the GPs for each of them. After this, we run through the process of removing any implausible points and selecting the points to sample in the next round in a very similar way to the first half of this project ([Spence 2025](#)). We iterate this process until the optimal point(s) are found.

5 Analysis of Results

The convergence tests I ran after augmenting the original method in Spence ([2025](#)) with different acquisition functions confirmed that my method still led to convergence to the correct point identified in the paper for every acquisition function. The convergence and speed tests can be found in the code in [Section 7](#).

From running our simulation for the shortcut model from the second half of the project on the Viking HPC, we were able to obtain some results ([York, n.d.](#)). The shortcut model initially gave us a long list of parameters, each with $F_{cod} = 0.42$ and $F_{had} \geq 0.34$. As $F_{had} = 0.34$ was given as an optimal parameter in the list, it means that this results in the same long-term catch of haddock as if we were to fish at any $F_{had} > 0.34$, as we were maximising the total long-term catch. Remembering that cod is the choke stock, this list also suggests that when we fish at $F_{cod} = 0.42$ this causes us to fish at $F_{had} = 0.34$ due to the ratio of cod to haddock that we are catching across the fleets ([Pace 2024](#)). Thus, $F_{cod} = 0.42$ and $F_{had} = 0.34$ are the optimal parameters suggested by the shortcut simulation.

These appear to be relatively high. ICES documents regularly suggest lower values for F_{MSY} in the North Sea region for cod ([ICES 2019c](#)), ([ICES 2020a](#)). Looking at the most recent advice, the model's output of $F_{cod} = 0.42$ is significantly above the $F_{MSY} = 0.231$ but below $F_{PA} = 0.464$ ([ICES 2025b](#)). This value aligns with my risk calculation by being below F_{PA} , as this ensures that [Equation 14](#) holds in the long-term ([ICES 2021b](#)). However, I would have expected my value to be closer to F_{MSY} as this ensures [Equation 14](#) and is maximising the long-term catch ([ICES 2021b](#)).

On the other hand, for Celtic Sea haddock the optimal parameter given by the shortcut model $F_{had} = 0.34$ is below the $F_{MSY} = 0.353$ and the $F_{PA} = 0.708$ given in the latest advice and sits in a similar area historically ([ICES 2025a](#)), ([ICES 2019b](#)), ([ICES 2020b](#)). This shows that the shortcut model has produced a safe value for F_{had} for Celtic Sea haddock according to the latest ICES guidelines. The shortcut model took 10 minutes and 34 seconds to run on the Viking HPCs ([York, n.d.](#)).

From running our simulation based on the fixed fishing mortality example on the MixME wiki we receive similar results, despite using the same dataset. Again, we obtain a list of parameters and are able to determine, because cod is the choke stock, that $F_{cod} = 0.38$ and $F_{had} = 0.28$ are the optimal parameters ([Pace 2024](#)). The F_{cod} falls slightly closer to the F_{MSY} value recommended by ICES and is still below F_{PA} ([ICES 2025b](#)). The F_{had} is again below both the F_{MSY} and the F_{PA} , suggesting haddock is being choked by cod ([ICES 2025a](#)). The simulation based on the fixed fishing

mortality example took 7 minutes and 48 seconds to run on the Viking HPCs, which is less time than the shortcut model. This could be in part due to varying usage levels on Viking (York, n.d.).

The values from this model are likely to be different to those from the shortcut model because they have been set up in different way. For example, one change is that the SSB is taken without including the pre-calculated noise in the shortcut model.

The discrepancy in the F_{cod} given by the models and those suggested by ICES (2025b) could be due to ICES maximising the long-term single-stock catch for cod whilst my simulations aim to maximise the long-term total catch for both cod and haddock (Sun et al. 2023). This could lead to the quota for cod being raised, whilst remaining precautionary, so that we can fish more haddock. Allowing this could result in a lower long-term catch for cod, but this reduction in cod catch could be made up for by the increase in haddock catch. Thus, the models may be trying to balance the trade off of reducing the long term catch of cod by fishing above the F_{MSY} for cod with catching more haddock if it does this (Pace et al. 2025a),(Sun et al. 2023). This would result in a value between F_{MSY} and F_{PA} for cod which is exactly what we see above. There are many other reasons that the model could be different from those conducted by ICES, but we will not focus on this here.

I then conducted a grid search of the whole parameter space. From this, I was able to find that the optimal parameters correspond to those identified above by the framework. These results are available from [Section 7](#).

6 Summary and Further areas for Development

We have created a framework that is able to speed up management strategy evaluation grid searches in single stock or multi-stock contexts, significantly reducing the burden of expensive runs of the objective function (Spence 2025). This was done by using advanced statistical techniques such as BHM and Gaussian Process Regression as well as providing some of our own insights by using acquisition functions and k-means clustering. We have compared the usefulness of different acquisition functions and decided upon one to implement in our framework. We have also thoroughly explained how the addition of k-means clustering improves the trade off between exploration and exploitation when sampling multiple points (Azimi, Fern, and Fern 2010),(Hartigan and Wong 1979). In addition, we have successfully demonstrated the usefulness of this framework by showing it takes less evaluations than in Spence (2025) to converge on the optimal point for the Case Study in [Section 5](#) and by applying our method to a dataset with multiple stocks.

Overall, this framework presents a more computationally efficient way for stakeholders to be able determine management strategies that will keep stocks healthy whilst maximising the long-term catch of the fishery (ICES 2019a). This is especially relevant at this time because many mixed fisheries are struggling after the collapse of an individual stock causes them to have to severely limit the catch of other stocks despite growing demand from consumers (ICES 2025c),(ICES 2025d), (United Nations. Fisheries Department 2024). By reducing computation time for MSE grid searches, we make it easier for more simulations to be run, which could allow fisheries advice to be published in a more timely manner (Spence 2025),(Jones, Schonlau, and Welch 1998). This could allow

organisations such as ICES to be able to keep pace with the changes in environment and in interactions between species (Holsman et al. 2019). As a result, fisheries advice could be more accurate and relevant, contributing towards a healthier ecosystem and planet.

6.1 Further development

One prominent area for further development would be moving on to using the dataset seen in the Proof of Concept section in Pace et al. (2025a). This is a realistic scenario based in the Celtic Sea and would help demonstrate the applicability of my method in more complex scenarios (Pace et al. 2025a). It would also be an example of my method in higher dimensions as there would now be three stocks (cod, haddock and whiting) and so I would be looking to find the optimum F_{target} for each. This means I would be optimising three variables as well as the catch and so it will be advisable to check if my method is still as comparatively fast in higher dimensions.

Furthermore, as there are 1000 iterations in this dataset, I will then be able to do the risk calculation as described in Pace et al. (2025a) and as is it is typically done across the industry (Pace et al. 2025b),(ICES 2019a). Once we have multiple iterations, we can follow the MixME definition that risk is the proportion of iterations in each year where $SSB < B_{lim}$, which is equivalent to $P(SSB < B_{lim})$ where the probability is taken over the number of iterations (Pace 2024).

Another area for improvement would be considering other metrics to optimise. For example, instead of maximising for long-term median catch, I could aim for Maximum Economic Yield which also takes into account the effort expended by the fleet to catch the stock (Sun et al. 2023). We could also consider the change in TAC (ICES 2021b). It is common for fisheries to want a relatively stable TAC as this makes the industry more predictable (Trusts 2019). This is occasionally taken into consideration by ICES (ICES 2021b).

Trophic interactions (interactions between species in a mixed fishery) are not considered in the MixME model (Pace et al. 2025a). However, there is evidence to suggest that these are still important factors even in areas dominated by fishing activity. This means we should be providing data about these relationships to our model as well to provide better projections into the future (Sun et al. 2023).

A last thing to consider would be modeling the uncertainty of the SSB values outputted by the MixME simulation and including this uncertainty in our GPs. MixME gives us the distribution that is modelling the pre-computed uncertainty in the observation error model object (Pace et al. 2025a), (Pace et al. 2025b). We would then include the standard deviation of the uncertainty for the SSB values in the GP as an appropriate input (provided by the Dicekriging package) (Roustant 2025). Then, the GP would take this into account when modelling SSB and so we would be able to estimate the SSB value for each point in the grid from the GP as normal (Roustant 2025). It is important to note that the uncertainty for the SSB values means we do not know that the value we get when we sample the SSB at a point is the true value of the SSB at that point and so we may want to sample the same point again to become more confident in what the true SSB value is at that point. The EI acquisition function does not allow this, as the EI of a point becomes zero once

you have sampled it (Frazier 2018). Hence, if we make sure we are using an acquisition function that can model sampling noise such as AEI or KG, we can continue the framework as beforehand, whilst accounting for the uncertainty of the SSB values (Huang et al. 2006),(Frazier 2018). The same process can be done for the uncertainty of the catch values outputted by the MixME model (Pace et al. 2025b).

7 Appendix

The code is all contained in the DissertationWork repository on Github (Bowen 2026). Please note that throughout this repository I have been allowed by my supervisor to use any AI tools which may be useful.

The visualisations of the GPs from the first half of the project can be seen in the Gaussian Process Visualisation and Experimentation folder in the `case_study8_3D.qmd` file.

The test for which acquisition function is best can be seen as the most recent tests in the “Actions” tab or can be seen in the Documents folder in the Deciding which acquisition function is best based on results from tests pdf. These include the speed and convergence tests.

The original MixME scripts can be found in the /MixME_experimentation/Original scripts folder.

The results of the grid search are available in the Grid_Search_Results folder.

The logs for my runs of the shortcut model and the model based on the fixed fishing mortality example can be found in the run_results_Optimising_ftarget_in_MixME_mult_points_parallel and run_results_Optimising_ftarget_in_MixME_mult_points_parallel folder respectively.

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